

Machine Learning 1

Mock Exam A

Duration: 2 hours Total: 100 points

Name: _____

Student ID: _____

Instructions

- Show all important intermediate steps. Answers without justification can receive reduced credit.
- State any additional assumptions that you use.
- Give matrix and vector dimensions where the question asks for them.
- This course uses numerator layout: if $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$, then its Jacobian has shape $n \times m$.
- Unless a question states otherwise, vectors are column vectors.
- The suggested working times total 110 minutes. Reserve about 10 minutes to check your answers.

Exercise 1: Multivariate derivatives and regularization 20 points, 18 minutes

1.1 Temperature-scaled log-softmax

Let $x \in \mathbb{R}^d$ and let $\tau > 0$. Define $s : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $\ell : \mathbb{R}^d \rightarrow \mathbb{R}^d$ by

$$s_i(x) = \frac{\exp(x_i/\tau)}{\sum_{k=1}^d \exp(x_k/\tau)}, \quad \ell_i(x) = \log s_i(x).$$

1. Derive $\partial \ell_i / \partial x_j$ using index notation. Express the answer using s_j , τ , and the Kronecker delta δ_{ij} . [5 points]
2. Write the full Jacobian $J_\ell(x)$ in matrix notation and state its dimensions. [4 points]
3. Show that $J_\ell(x)\mathbf{1} = 0$, where $\mathbf{1}$ is the all-ones vector. Explain this result using a transformation of x that leaves $\ell(x)$ unchanged. [4 points]

1.2 Generalized ridge regression

Let $X \in \mathbb{R}^{n \times p}$, $y \in \mathbb{R}^n$, $\theta \in \mathbb{R}^p$, and $\gamma > 0$. Let $R \in \mathbb{R}^{p \times p}$ be symmetric positive definite. Consider

$$L(\theta) = \frac{1}{n} \|X\theta - y\|_2^2 + \gamma \theta^\top R \theta.$$

4. Derive the stationary point of $L(\theta)$. Then show that it is the unique global minimizer, even when X does not have full column rank. [7 points]

Exercise 2: Waiting times and Bayesian estimation 24 points, 25 minutes

A monitoring system checks a component repeatedly. Each check succeeds independently with probability q , where $0 < q < 1$. Let K be the number of failed checks before the first success. Its probability mass function is

$$p(K = k | q) = q(1 - q)^k, \quad k = 0, 1, 2, \dots$$

You may use

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1 - r}, \quad |r| < 1.$$

1. Derive $\mathbb{E}[K | q]$. Do not only state the result. **[3 points]**
2. Compute $P(K = 2 | q = 0.25)$. Give an exact or decimal value. **[3 points]**

Now suppose that $K_1, \dots, K_N$ are independent observations and define $S = \sum_{n=1}^N K_n$. Assume $S > 0$.

3. Write the likelihood and log-likelihood of q , up to constants that do not depend on q . **[3 points]**
4. Derive the maximum-likelihood estimator of q . Verify that your stationary point is a maximum. **[4 points]**

Place a Beta(a, b) prior on q , with density

$$p(q) \propto q^{a-1}(1 - q)^{b-1}, \quad 0 < q < 1,$$

where $a > 1$ and $b > 1$.

5. Derive the posterior distribution of q and identify its parameters. **[4 points]**
6. Derive the MAP estimator of q . **[4 points]**
7. Identify a one-dimensional statistic of the data that contains all data dependence in the posterior. Briefly justify your answer. **[3 points]**

Exercise 3: Estimating an unknown range 14 points, 14 minutes

Measurements $X_1, \dots, X_N$ are independent and uniformly distributed on $[-M, M]$, where $M > 0$ is unknown. Define

$$A = \max_{1 \leq n \leq N} |X_n|.$$

Assume that at least one observed value is nonzero, so $A > 0$.

1. Write the likelihood $p(x_1, \dots, x_N | M)$, including the support constraint. **[4 points]**
2. Derive the maximum-likelihood estimator of M . For the observations

$$-3.2, \quad 1.1, \quad 2.4, \quad -2.8, \quad 0.5,$$

give its numerical value.

[3 points]

You may use

$$\mathbb{E}[A] = \frac{NM}{N+1}, \quad \mathbb{E}[A^2] = \frac{NM^2}{N+2}.$$

3. Derive a multiplicative correction of the maximum-likelihood estimator that is unbiased for M .
[3 points]
4. Derive the variance of your unbiased estimator and simplify the result. [4 points]

Exercise 4: Multi-output regression with correlated noise 23 points, 29 minutes

For each observation, let $\phi_n \in \mathbb{R}^m$ be a feature vector and $t_n \in \mathbb{R}^k$ be a target vector. Let $W \in \mathbb{R}^{m \times k}$. The model is

$$t_n = W^\top \phi_n + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \sigma^2 C),$$

where $C \in \mathbb{R}^{k \times k}$ is known and symmetric positive definite, while $\sigma^2 > 0$ is unknown. The noise vectors are independent.

Let $\Phi \in \mathbb{R}^{N \times m}$ have rows $\phi_n^\top$, and let $T \in \mathbb{R}^{N \times k}$ have rows $t_n^\top$. Assume that Φ has full column rank, $N > m$, and the data cannot be fitted exactly.

1. State the dimensions of ΦW , $T - \Phi W$, and $(T - \Phi W)^\top (T - \Phi W)$. [2 points]
2. Derive the log-likelihood of W and σ^2 , including all terms that depend on either parameter. You may express the quadratic term using a trace. [6 points]
3. Derive the maximum-likelihood estimator of W . Show why it does not depend on C or σ^2 . [7 points]
4. Using the fitted value of W , derive the maximum-likelihood estimator of σ^2 . [5 points]
5. Let $E = T - \Phi W_{\text{ML}}$. Show that $\Phi^\top E = 0$ and explain the geometric meaning of this identity. [3 points]

Exercise 5: Information from one Bayesian observation 19 points, 24 minutes

After observing a data set $\mathcal{D}_N$, the posterior over the regression weights is

$$p(w \mid \mathcal{D}_N) = \mathcal{N}(w \mid \mu_N, \Sigma_N),$$

where $w, \mu_N \in \mathbb{R}^m$ and $\Sigma_N \in \mathbb{R}^{m \times m}$ is symmetric positive definite. A new observation has feature vector $\phi \in \mathbb{R}^m$, target $t \in \mathbb{R}$, and likelihood

$$p(t \mid w, \phi) = \mathcal{N}(t \mid w^\top \phi, \beta^{-1}),$$

where $\beta > 0$ is the observation-noise precision.

The standard precision and natural-parameter updates are

$$\begin{aligned} \Sigma_{N+1}^{-1} &= \Sigma_N^{-1} + \beta \phi \phi^\top, \\ \Sigma_{N+1}^{-1} \mu_{N+1} &= \Sigma_N^{-1} \mu_N + \beta \phi t. \end{aligned}$$

You may use the Sherman–Morrison identity

$$(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1}uv^\top A^{-1}}{1 + v^\top A^{-1}u}.$$

1. Derive the rank-one covariance update

$$\Sigma_{N+1} = \Sigma_N - \frac{\Sigma_N \phi \phi^\top \Sigma_N}{\beta^{-1} + \phi^\top \Sigma_N \phi}.$$

[4 points]

2. Derive the innovation form of the mean update

$$\mu_{N+1} = \mu_N + K(t - \phi^\top \mu_N),$$

and give the gain vector K . Explain the roles of the direction and the scalar residual in this update. [5 points]

3. Before observing t , derive the posterior predictive distribution $p(t \mid \phi, \mathcal{D}_N)$. Give both its mean and variance, and identify the two sources of predictive variance. [4 points]

4. For any direction $a \in \mathbb{R}^m$, derive

$$a^\top (\Sigma_N - \Sigma_{N+1}) a.$$

Show that it is nonnegative. State exactly when it is zero and explain what this says about the parameter directions learned from the observation. [6 points]

End of Mock Exam A

Machine Learning 1

Mock Exam B

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Exercise 1: Normalization and a nonsmooth loss — 20 points, 20 minutes

1.1 Vector normalization

Define $h : \mathbb{R}^d \setminus \{0\} \rightarrow \mathbb{R}^d$ by

$$h(x) = \frac{x}{\|x\|_2}.$$

1. Derive $\partial h_i / \partial x_j$ using index notation. [5 points]
2. Write the full Jacobian $J_h(x)$ in matrix notation and state its dimensions. [4 points]
3. Show that $J_h(x)x = 0$. Explain why this agrees with the behavior of $h(cx)$ for $c > 0$. [3 points]

1.2 Absolute-error regression

Let $a, w \in \mathbb{R}^p$, with $a \neq 0$, let $y \in \mathbb{R}$, and let $\lambda > 0$. Define

$$f(w) = |a^\top w - y| + \frac{\lambda}{2} w^\top w.$$

4. When $a^\top w - y \neq 0$, derive the $1 \times p$ derivative of $f(w)$ using the course's numerator-layout convention. [4 points]
5. State why the derivative does not exist when $a^\top w - y = 0$. Give one valid subgradient at such a point, and write one subgradient-descent update with step size $\eta > 0$. [4 points]

Exercise 2: A transformed waiting-time model 24 points, 25 minutes

Let X have density

$$p(x | \lambda) = \begin{cases} r\lambda x^{r-1} \exp(-\lambda x^r), & x > 0, \\ 0, & x \leq 0, \end{cases}$$

where $r > 0$ is known and $\lambda > 0$ is unknown.

1. Show that this is a valid probability density. [3 points]
2. Define $Y = X^r$. Derive the density of Y , identify its distribution, and hence find $\mathbb{E}[X^r | \lambda]$. [4 points]
3. Derive $P(X > c | \lambda)$ for $c \geq 0$. Evaluate it for $r = 2$, $\lambda = 0.25$, and $c = 2$. [3 points]

Now let $X_1, \dots, X_N$ be independent observations and define $S = \sum_{n=1}^N X_n^r$. Assume $S > 0$.

4. Derive the maximum-likelihood estimator of λ . Verify that it maximizes the likelihood. [4 points]

Use a Gamma prior with shape a and rate b :

$$p(\lambda) \propto \lambda^{a-1} \exp(-b\lambda),$$

where $a > 0$ and $b > 0$.

5. Derive the posterior distribution and identify its parameters. [4 points]
6. Derive the MAP estimator. [3 points]
7. Give the posterior mean and compare it with the MAP estimator. What happens to their difference as N increases while S/N remains fixed and positive? [3 points]

Exercise 3: Bias, variance, and shrinkage 15 points, 16 minutes

Let $X_1, \dots, X_N$ be independent random variables with

$$\mathbb{E}[X_n] = \mu, \quad \text{Var}(X_n) = \sigma^2,$$

where the known value σ^2 is strictly positive. Let m_0 be a fixed reference value and define, for $0 \leq c \leq 1$,

$$\tilde{\mu}_c = c\bar{X} + (1 - c)m_0.$$

1. Derive the bias of $\tilde{\mu}_c$ as an estimator of μ . State when it is unbiased. [3 points]
2. Derive its variance. [3 points]
3. Derive its mean squared error. You may use $\text{MSE} = \text{variance} + \text{bias}^2$. [3 points]
4. For fixed μ , find the value of $c \in [0, 1]$ that minimizes the mean squared error. [4 points]
5. Explain why the minimizing value from part 4 is called an oracle choice. Describe its limiting behavior as $N \rightarrow \infty$, distinguishing $\mu = m_0$ from $\mu \neq m_0$. [2 points]

Exercise 4: Multi-output MAP regression

22 points, 24 minutes

Let $T \in \mathbb{R}^{N \times k}$, $\Phi \in \mathbb{R}^{N \times m}$, and $W \in \mathbb{R}^{m \times k}$. Consider the likelihood

$$p(T | W) \propto \exp\left(-\frac{\beta}{2} \|T - \Phi W\|_F^2\right),$$

where $\beta > 0$. Let $M_0 \in \mathbb{R}^{m \times k}$ and let $R \in \mathbb{R}^{m \times m}$ be symmetric positive definite. Use the prior

$$p(W) \propto \exp\left(-\frac{\alpha}{2} \text{tr}((W - M_0)^\top R (W - M_0))\right),$$

where $\alpha > 0$.

1. Write the negative log-posterior, up to additive constants independent of W . **[4 points]**
2. Derive the MAP estimator of W . Define $\lambda = \alpha/\beta$ and give your final answer in terms of λ . **[7 points]**
3. Show that the MAP estimator is unique, even if Φ does not have full column rank. **[3 points]**
4. Describe the limits of W_{MAP} as $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$. State any rank assumption needed for the first limit. **[4 points]**
5. As λ increases, describe the changes in $\|T - \Phi W_{\text{MAP}}\|_F^2$ and in

$$\text{tr}((W_{\text{MAP}} - M_0)^\top R (W_{\text{MAP}} - M_0)).$$

Under the usual correctly specified regression assumptions, explain the general effect of stronger shrinkage on variance and bias, including the special case in which M_0 equals the true coefficient matrix. **[4 points]**

Exercise 5: Two sequential Bayesian updates

19 points, 25 minutes

The current posterior in a Bayesian linear regression model is

$$p(w | \mathcal{D}_N) = \mathcal{N}(w | \mu_N, \Sigma_N).$$

Assume that Σ_N is symmetric positive definite. Two new observations are

$$(\phi_a, t_a, \beta_a) \quad \text{and} \quad (\phi_b, t_b, \beta_b),$$

where each ϕ is in $\mathbb{R}^m$, each target is scalar, and each positive β is the precision of that observation's Gaussian noise. Define the current natural parameter

$$\eta_N = \Sigma_N^{-1} \mu_N.$$

1. Derive the posterior precision after incorporating both observations. **[5 points]**
2. Derive the final natural parameter and use it to give the final posterior mean. **[4 points]**
3. Use your answers to show that the final posterior does not depend on the order in which the two observations are processed. State the probabilistic assumption that makes this result valid. **[4 points]**
4. Suppose $\phi_b = \phi_a$. Show that the two observations have the same effect on the posterior as one observation with precision $\beta_a + \beta_b$. Derive the equivalent target value. **[4 points]**

5. If $\phi_b = c\phi_a$ for a nonzero scalar c , does the second observation provide information in a new parameter-space direction? Explain briefly. **[2 points]**

End of Mock Exam B